

Project Dataset

The Week 2 project uses a synthetic enterprise sales transaction dataset containing **500,000 transaction records**.

The required dataset file is:

`sales_transactions.csv`

The dataset contains the following columns:

- TransactionID
- CustomerID
- CustomerName
- Gender
- Age
- ProductID
- ProductName
- Category
- Quantity
- UnitPrice
- Discount
- GST
- PaymentMethod
- City
- State
- OrderDate
- SalesPerson

Dataset File Size

Due to the large number of transaction records, the generated CSV file is larger than the standard file size limit for direct browser uploads to GitHub.

Therefore, the dataset is generated programmatically using R instead of directly uploading the complete CSV file to the GitHub repository.

Dataset Generation

The dataset can be generated by running the following R script:

```
Task1_Generate_Dataset.R
```

The script creates a synthetic dataset containing at least 500,000 transaction records and saves it as:

```
sales_transactions.csv
```

The following command is used to save the dataset:

```
fwrite(  
  sales_transactions,  
  "sales_transactions.csv"  
)
```

Steps to Generate the Dataset

Step 1: Install R and RStudio

Install R and RStudio on your computer.

Step 2: Install the Required Package

Run the following command:

```
install.packages("data.table")
```

Step 3: Open the Project

Open the project folder in RStudio.

Step 4: Run the Dataset Generation Script

Run:

```
Task1_Generate_Dataset.R
```

Step 5: Dataset Creation

After successfully running the script, the following file will be created:

```
sales_transactions.csv
```

The generated dataset contains:

```
Rows: 500,000  
Columns: 17
```

Loading the Dataset

After generating the CSV file, it can be loaded using the `fread()` function:

```
library(data.table)

sales_transactions <- fread(
  "sales_transactions.csv"
)
```

The dataset can then be used for all subsequent project tasks.

Project Workflow

The complete project workflow is:

```
Generate Dataset
  ↓
Load Dataset using fread()
  ↓
Explore Dataset
  ↓
Clean and Prepare Data
  ↓
Perform High-Speed Data Manipulation
  ↓
Create Columns using :=
  ↓
Create Indexes using setkey()
  ↓
Perform Grouping and Aggregation
  ↓
Perform Business Analysis
  ↓
Export Business Reports
  ↓
Generate Visualizations
```

Reason for Not Directly Uploading the CSV File

The dataset contains 500,000 transaction records and is therefore relatively large in size.

Instead of reducing the number of records or changing the required dataset structure, the dataset is generated dynamically using R.

This approach has the following advantages:

- Maintains the required 500,000 transaction records.
- Preserves the required CSV file name.
- Allows the dataset to be reproduced at any time.
- Reduces the size of the GitHub repository.
- Demonstrates synthetic dataset generation using R.
- Makes the project easier to reproduce.

Reproducibility

The dataset generation script uses:

```
set.seed(123)
```

This ensures that the same synthetic dataset can be reproduced whenever the program is executed again.

Therefore, anyone with the project code can generate the required:

```
sales_transactions.csv
```

dataset by running the dataset generation script.

Conclusion

Although the generated dataset is large, the complete dataset-generation process is included in the project source code.

The dataset can be recreated at any time by running the R script. This maintains the project requirements while avoiding unnecessary repository size limitations.

The generated file is named:

```
sales_transactions.csv
```

and contains at least:

```
500,000 transaction records
```

with all required enterprise sales transaction attributes.